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Unit 7: Emerging Technologies — Complete Question Bank

Multiple Choice Questions

Section A: Artificial Intelligence

1. What does Artificial Intelligence simulate?

- a) Human physical abilities
 - b) Human thinking and learning ability in computer systems**
 - c) Human emotional responses only
 - d) Human communication patterns only
-

2. Which of the following is NOT a subfield of AI?

- a) Machine Learning
 - b) Natural Language Processing
 - c) Computer Vision
 - d) DBMS**
-

3. The first AI program, the Logic Theorist, was created in:

- a) 1945
 - b) 1965
 - c) 1955**
 - d) 1975
-

4. Who created the first AI program, the Logic Theorist?

- a) Bill Gates and Steve Jobs
 - b) Allen Newell and Herbert A. Simon**
 - c) Kevin Ashton and Tim Berners-Lee
 - d) Alan Turing and John McCarthy
-

5. Which AI subfield helps computers understand and respond in human language?

- a) Computer Vision
 - b) Robotics
 - c) Natural Language Processing (NLP)**
 - d) Deep Learning
-

6. Which of the following is an example of NLP in everyday use?

- a) A robot vacuum cleaner
 - b) Siri or Alexa understanding your questions**
 - c) A self-driving car detecting obstacles
 - d) A smart thermostat adjusting temperature
-

7. Deep Learning uses complex structures called:

- a) Neural Networks**
 - b) Decision Trees
 - c) Relational Databases
 - d) Binary Search Trees
-

8. Neural networks are inspired by:

- a) Computer processors
 - b) The human brain**
 - c) The internet structure
 - d) Mathematical equations
-

9. Machine Learning is best described as:

- a) Programming computers with fixed rules
 - b) Teaching computers to learn from experience without explicit programming**
 - c) Building robots that mimic humans
 - d) Creating databases for AI systems
-

10. Which AI subfield enables computers to interpret and understand images and videos?

- a) NLP
 - b) Robotics
 - c) Computer Vision**
 - d) Machine Learning
-

11. Self-driving cars primarily rely on which technology?

- a) IoT only
 - b) NLP only
 - c) AI**
 - d) Cloud computing only
-

12. Which of the following is an application of AI in healthcare?

- a) Personalized drug development
 - b) Automated diagnosis
 - c) Remote patient monitoring
 - d) All of the above**
-

13. AI in finance is used for:

- a) Fraud detection
 - b) Algorithmic trading
 - c) Personalized investment recommendations
 - d) All of the above**
-

14. AI is used in agriculture for:

- a) Predictive analytics for crop yields
 - b) Automated irrigation systems
 - c) Computer vision for disease and pest detection
 - d) All of the above**
-

15. In social media, AI is used for:

- a) Personalized content recommendations
 - b) Sentiment analysis
 - c) Targeted advertising
 - d) All of the above**
-

16. Which of the following best describes Explainable (Whitebox) AI algorithms?

- a) The decision-making process is transparent and understandable**
 - b) The decision-making process is hidden and complex
 - c) They always use neural networks
 - d) They perform better than blackbox algorithms
-

17. Which of the following is an example of a Blackbox (Unexplainable) AI algorithm?

- a) Simple decision tree
 - b) Linear regression
 - c) Deep learning neural network**
 - d) Rule-based expert system
-

18. The key difference between whitebox and blackbox AI models is:

- a) The complexity of the model
 - b) The ability to understand the decision-making process**
 - c) The performance of the model
 - d) The training data used
-

19. Google's AlphaGo used which type of AI technique?

- a) Supervised Learning
 - b) Natural Language Processing
 - c) Reinforcement Learning**
 - d) Computer Vision
-

20. What is the primary purpose of using AI techniques in machine learning models?

- a) To improve accuracy
 - b) To enhance interpretability
 - c) To reduce computational complexity
 - d) All of the above**
-

21. AI in e-commerce is used for:

- a) Personalized product recommendations
 - b) Intelligent chatbots for customer support
 - c) Fraud detection systems
 - d) All of the above**
-

22. Robotics is defined as:

- a) The study of computer algorithms
 - b) The science of building and programming robots**
 - c) A branch of NLP
 - d) A type of machine learning
-

Section B: Internet of Things (IoT)

23. The term "Internet of Things" was coined by:

- a) Allen Newell
 - b) Tim Berners-Lee
 - c) Kevin Ashton**
 - d) Herbert A. Simon
-

24. When was the term "Internet of Things" coined?

- a) 1985
 - b) 2005
 - c) 1999**
 - d) 2010
-

25. IoT is defined as:

- a) A network of AI-powered robots
 - b) A network of physical objects equipped with sensors and software to exchange data over the internet**
 - c) A type of cloud computing service
 - d) A cybersecurity framework
-

26. Which of the following is NOT a component of an IoT system?

- a) Sensors
 - b) Actuators
 - c) Compiler**
 - d) Networks
-

27. What is the role of sensors in an IoT system?

- a) To detect and measure physical properties like temperature, humidity, and motion**
 - b) To connect devices to the internet
 - c) To analyse and process data
 - d) To convert energy into motion
-

28. What is the function of actuators in an IoT system?

- a) To collect data from the environment
 - b) To convert energy into motion and act on data to generate output**
 - c) To provide communication pathways
 - d) To analyse collected data
-

29. Networks in IoT systems serve the purpose of:

- a) Detecting physical properties
 - b) Processing data analysis
 - c) Providing communication pathways connecting sensors and devices to the internet**
 - d) Converting energy into motion
-

30. In 2020, how many IoT devices were in use worldwide?

- a) 5 billion
 - b) 10 billion
 - c) Over 20 billion**
 - d) 50 billion
-

31. Which of the following is a security concern in IoT deployments?

- a) Device vulnerability
 - b) Data privacy
 - c) Lack of standardization
 - d) All of the above**
-

32. Which of the following is a recommended security measure for IoT devices?

- a) Using simple passwords for convenience
 - b) Regular software and firmware updates**
 - c) Sharing passwords across devices
 - d) Avoiding encryption to speed up data transfer
-

33. Which of the following is an application of IoT in the transportation domain?

- a) Smart traffic management
 - b) Vehicle-to-Vehicle (V2V) communication
 - c) Predictive maintenance of vehicles
 - d) All of the above**
-

34. IoT in healthcare is used for:

- a) Tracking vital signs
 - b) Reminding patients to take medication
 - c) Alerting healthcare providers in emergencies
 - d) All of the above**
-

35. A smart home system is an example of IoT because:

- a) It uses AI to make decisions
 - b) It connects internet-enabled appliances like temperature controls, lighting, and cameras**
 - c) It stores large amounts of data in the cloud
 - d) It uses NLP to communicate with users
-

36. Data in IoT systems can be analysed:

- a) Only on the device itself
 - b) Only in the cloud
 - c) Only on a central server
 - d) On the device, in the cloud, or on a central server**
-

37. Encryption in IoT systems is used to:

- a) Speed up data transmission
 - b) Reduce power consumption
 - c) Protect data from being intercepted by hackers**
 - d) Increase device connectivity
-

38. IoT is significant because it allows for:

- a) Replacing human workers with machines
 - b) Seamless integration of the physical and digital worlds**
 - c) Unlimited data storage at no cost
 - d) Eliminating the need for cybersecurity
-

39. Which of the following best describes "Data Analysis" as an IoT component?

- a) Processing and analysing collected sensor data to gain insights and make decisions**
 - b) Detecting physical properties in the environment
 - c) Providing communication pathways between devices
 - d) Converting electrical signals into motion
-

40. Strong passwords are recommended for IoT devices in order to:

- a) Speed up the device
 - b) Prevent unauthorized access**
 - c) Improve data analysis
 - d) Enhance device connectivity
-

Short Questions

Section A: Artificial Intelligence

1. Define Artificial Intelligence (AI).
 2. Who created the first AI program and what was it called?
 3. What is Machine Learning? How is it different from traditional programming?
 4. What is Deep Learning and what structures does it use?
 5. Define Natural Language Processing (NLP) and give two real-world examples.
 6. What is Computer Vision? Give one practical application.
 7. Define Robotics as a subfield of AI.
 8. List any four domains where AI is applied and briefly describe its role in each.
 9. How is AI used in the healthcare sector?
 10. How does AI contribute to the agriculture sector?
 11. What are the two broad categories of AI algorithms based on interpretability?
 12. What are Explainable (Whitebox) AI algorithms? Give one example.
 13. What are Unexplainable (Blackbox) AI algorithms? Give one example.
 14. What is the key difference between whitebox and blackbox AI models?
 15. What is Reinforcement Learning? Give an example of its application.
 16. How is AI used in the finance industry?
 17. How is AI used in social media platforms?
 18. How is AI transforming the automobile industry?
 19. Explain the role of AI techniques in advancing machine learning models.
 20. Give two examples of AI applications in e-commerce.
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Section B: Internet of Things (IoT)

1. Define the Internet of Things (IoT).
2. Who coined the term "Internet of Things" and when?
3. What is the significance of IoT in connecting devices and systems?
4. List and briefly describe the five main components of an IoT system.
5. What is the role of sensors in an IoT system?
6. What is the role of actuators in an IoT system? How are they different from sensors?
7. What is the role of networks in an IoT system?
8. Explain how data analysis works within an IoT system.
9. Give one example of a smart home IoT application and describe how it works.
10. Describe applications of IoT in the healthcare domain.
11. Describe applications of IoT in the transportation domain.
12. What security and privacy concerns arise from IoT deployments?
13. List three security measures that should be taken to protect IoT devices.
14. Why is encryption important in IoT systems?
15. Why is it important to regularly update the software and firmware of IoT devices?
16. How does IoT contribute to improving efficiency in smart homes?
17. How does IoT benefit the healthcare sector in terms of patient monitoring?
18. What is meant by "Vehicle-to-Vehicle (V2V) communication" in the context of IoT?
19. Explain the importance of using strong passwords for IoT devices.
20. How does IoT enable the seamless integration of the physical and digital worlds?

Tip for Students: For MCQs, focus on definitions, examples, and the distinguishing features of each concept. For short questions, always structure your answer with a clear definition followed by an example where possible.